Zingiber officinale Ameliorates Oxidative Stress in Adult Wistar Rats with Cytoarchitectural Distortion of the Pituitary Gland

[*1]Olugbenga M.A., [2]Ekibade P.N., [3]Olabanji O.J., [1]Eromosele M.O., [3]Okosun M.

[1]*Department of Anatomy, Faculty of Basic Medical Sciences, College of Medicine, Ambrose Alli University, Ekpoma, Edo State, Nigeria.*

[2]*Department of Anatomy, School of Basic Medical Sciences, College of Medical Sciences, University of Benin, Benin City, Nigeria;*

[3]*Department of Anatomy, Chrisland University, Abeokuta, Ogun State, Nigeria.*

ABSTRACT

This study investigated the dose-dependent effects of *Zingiber officinale* (Ginger) on oxidative stress biomarkers and pituitary histoarchitecture in rats. Fifteen adult female Wistar rats were randomly assigned to three groups; control, 100 mg/kg and 300 mg/kg body weight of *Zingiber officinale* extract orally for 14 days. Body weights were monitored and at the end of treatment, blood samples were collected for the determination of serum malondialdehyde, catalase (CAT), reduced glutathione (GSH), and superoxide dismutase (SOD), while the pituitary glands were harvested for histological examination. Data were analysed using appropriate statistical methods, with statistical significance set at $p < 0.05$. Administration of *Zingiber officinale* resulted in a dose-dependent reduction in body weight gain and pituitary gland weight. Catalase activity increased in treated rats, whereas GSH and SOD activities declined relative to the control group. Malondialdehyde levels were lower in the treated groups, although differences were not statistically significant. Histological examination revealed varying degrees of cytoarchitectural distortion and inflammatory cell infiltration in the pituitary gland, with the most severe alterations occurring in rats administered 300 mg/kg extract. *Zingiber officinale* exhibited dose-dependent effects on oxidative stress biomarkers and pituitary histoarchitecture in adult female Wistar rats. While the extract enhanced catalase activity and showed evidence of reduced lipid peroxidation, the concomitant reductions in GSH and SOD activities together with histological alterations suggest that higher doses may adversely affect pituitary integrity. These findings highlight the importance of dose consideration when using *Zingiber officinale* for its therapeutic benefits.

Keywords: *Zingiber officinale*, Oxidative stress biomarkers, Pituitary gland, histoarchitecture, Antioxidants, Rats.

***Corresponding author. E-mail address:** maryolugbenga@aauekpoma.edu.ng; olugmary@gmail.com; Tel.: +2348066497690.

1. INTRODUCTION

Zingiber officinale (ginger) is a widely cultivated medicinal plant belonging to the family Zingiberaceae. It originated in South-East Asia and has since been widely used across the world as a spice, flavouring agent, and traditional herbal remedy. The rhizome contains a rich array of bioactive phytochemicals responsible for its diverse medicinal properties. According to Jolad *et al*. [1], the phytochemical constituents of *Zingiber officinale* are broadly classified into volatile and non-volatile compounds. The volatile constituents, primarily sesquiterpenes and monoterpenoids, are responsible for the characteristic aroma and flavour of ginger, whereas most of its biological activities are attributed to the non-volatile compounds, including gingerols, shogaols, paradols, and zingerone [2].

These bioactive compounds have been reported to possess anti-inflammatory, antioxidant,

antimicrobial, digestive, and metabolic properties. Consequently, ginger has been widely utilised in the management of arthritis, rheumatism, indigestion, peptic ulcers, hypertension, diabetes mellitus, and certain forms of cancer. Its phytochemicals have also been shown to modulate excessive production of reactive oxygen species (ROS), thereby reducing oxidative damage associated with ageing, neurodegenerative disorders, cardiovascular diseases, and other chronic conditions [3].

Oxidative stress occurs when the generation of reactive oxygen species exceeds the body's antioxidant defence capacity [4]. This imbalance results in oxidative damage to cellular lipids, proteins, and nucleic acids. Although ROS play essential physiological roles in cellular signalling and normal metabolic processes, excessive production leads to cellular dysfunction, tissue injury, and disease. Antus [5] described oxidative stress as an imbalance favouring oxidants over antioxidant defence mechanisms, resulting in molecular and cellular damage. Idrissi *et al*. [6] further explained that the balance between oxidants and antioxidants is influenced by the prevailing chemical environment, while Gulcin [7] defined antioxidants as substances capable of delaying or preventing oxidation, even at relatively low concentrations. Since the degree of oxidative stress varies among tissues and cellular compartments, understanding its role has become increasingly important in the pathogenesis of numerous diseases [8].

Because oxidative stress contributes to the development of ageing, cancer, diabetes mellitus, cardiovascular diseases, neurodegenerative disorders, and inflammatory conditions, considerable attention has been directed towards identifying safe and effective natural antioxidants [9]. *Zingiber officinale* has emerged as one of the most promising medicinal plants in this regard. Numerous studies have demonstrated that its bioactive constituents can scavenge free radicals, inhibit lipid peroxidation, and enhance endogenous antioxidant defence systems [10]. Among these compounds, 6-shogaol exhibits potent antioxidant and anti-inflammatory activities. Experimental studies have shown that ginger supplementation enhances glutathione levels and increases the activities of several antioxidant enzymes [11, 12]. Likewise, zingerone has been reported to inhibit lipid peroxidation and neutralise superoxide (O_2^-) and hydroxyl (•OH) radicals. These findings suggest that ginger may have therapeutic potential in preventing oxidative stress-related disorders, including neurodegenerative diseases such as Parkinson's disease [12].

Ginger has a long history of cultivation and utilisation worldwide. Although it originated in South-East Asia, it is now cultivated extensively in tropical and subtropical regions, including Nigeria, Sierra Leone, Indonesia, Bangladesh, Jamaica, Fiji, Haiti, and Mexico. Currently, India and China are the leading producers of ginger globally [13]. Beyond its culinary applications, ginger essential oil has demonstrated considerable antimicrobial, antifungal, antioxidant, and preservative properties, making it useful in preventing microbial spoilage and oxidative rancidity in food products [3].

The pituitary gland is a vital endocrine organ responsible for regulating growth, metabolism, reproduction, and numerous physiological processes through hormonal control. Often referred to as the "master gland" of the endocrine system, it maintains endocrine homeostasis by regulating the activities of several peripheral endocrine glands [14]. Normal pituitary function is therefore essential for maintaining hormonal balance and overall physiological health [15]. Disorders of the pituitary gland, including prolactinoma, acromegaly, Cushing's disease, non-functioning pituitary adenomas, and hypopituitarism, although relatively uncommon, are associated with significant morbidity. Emerging evidence suggests that pituitary structure and function may be

influenced by environmental factors, metabolic disturbances, and oxidative stress [16, 17].

Recent studies have also suggested that *Zingiber officinale* may influence endocrine function. Experimental evidence indicates that administration of ginger extract can increase circulating oxytocin levels in laboratory animals [10], raising the possibility that ginger may exert modulatory effects on pituitary function and hormone secretion. However, information regarding its effects on pituitary histoarchitecture and oxidative stress biomarkers remains limited. Given the central role of oxidative stress in disease pathogenesis and the growing evidence of the antioxidant and potential endocrine-modulating properties of *Zingiber officinale*, further investigation into its effects on the pituitary gland is warranted. Therefore, this study was designed to evaluate the dose-dependent effects of *Zingiber officinale* on oxidative stress biomarkers and pituitary histoarchitecture in adult female Wistar rats.

METHODS

2.1. Study Design: This study was an experimental animal investigation designed to evaluate the dose-dependent effects of *Zingiber officinale* on oxidative stress biomarkers and pituitary histoarchitecture in adult female Wistar rats. The experiment employed a controlled design involving graded doses of *Zingiber officinale* extract administered over a 14-day period under standard laboratory conditions.

2.2. Study setting: The experimental procedures were carried out in the Histology Laboratory and the Animal Research Facility of the Department of Anatomy, Ambrose Alli University, Ekpoma, Edo State, Nigeria.

2.3. Experimental animals: Fifteen healthy adult female Wistar rats, aged 14 weeks and weighing an average of 171 g, were used for this study. The animals were housed in well-ventilated wooden cages under standard laboratory conditions and allowed free access to commercial rat chow and clean drinking water *ad libitum*. They were acclimatised to the laboratory environment for seven days before commencement of the experiment.

Following acclimatisation, the animals were randomly allocated into three groups (n = 5 per group) as follows:
Group A (Control): Received standard rat chow and distilled water only for 14 days.
Group B: Received *Zingiber officinale* extract at a dose of 100 mg/kg body weight orally once daily for 14 days.
Group C: Received *Zingiber officinale* extract at a dose of 300 mg/kg body weight orally once daily for 14 days.

2.4. Preparation of *Zingiber officinale* extract: Fresh rhizomes of *Zingiber officinale* were thoroughly washed, sliced into small pieces, and sun-dried for several days before being milled into a fine powder. The powdered material was stored in airtight containers until extraction. Two hundred grams (200 g) of the powdered ginger were extracted with 100 mL of absolute ethanol using a Soxhlet extraction apparatus. The extract was concentrated and prepared at doses equivalent to 100 mg/kg and 300 mg/kg body weight for oral administration.

2.5. Experimental procedure: The initial body weights of all animals were recorded before treatment commenced and subsequently measured on Days 3, 6, 9, 12, and 14. The appropriate dose of *Zingiber officinale* extract was administered orally once daily throughout the experimental period.

On Day 14, the animals were humanely euthanised under appropriate laboratory procedures. The pituitary glands were carefully excised, rinsed in normal saline to remove blood contaminants, gently blotted dry with filter paper, and weighed using a sensitive electronic balance.

2.6. Sample collection and laboratory analysis: Following excision, the pituitary

glands were fixed in 10% neutral buffered formalin using a fixative-to-tissue ratio of 10:1 (v/w). The tissues were subsequently processed using standard histological techniques involving dehydration, clearing, paraffin embedding, sectioning, and haematoxylin and eosin (H&E) staining. The stained sections were examined microscopically for histoarchitectural alterations.

Blood samples were collected for the determination of oxidative stress biomarkers. Catalase (CAT) activity was determined using the method described by Sinha [18], lipid peroxidation was assessed by measuring thiobarbituric acid reactive substances (TBARS) according to Varshney and Kale [19], while superoxide dismutase (SOD) activity was determined using the method of Kono [20]. Reduced glutathione (GSH) levels were determined using an established standard protocol. All biochemical analyses were performed in duplicate to ensure analytical accuracy and reproducibility.

2.7. Statistical analysis: Data were expressed as mean ± standard error of the mean (SEM). Statistical analyses were performed using IBM SPSS Statistics version 20.0 (IBM Corp., Armonk, NY, USA). Comparisons among the three experimental groups were performed using one-way analysis of variance (ANOVA), followed by Tukey's post hoc multiple comparison test where appropriate. Statistical significance was accepted at $p < 0.05$.

3. RESULTS

3.1. Changes in body weight and pituitary gland weight

Table 1 presents the mean body weights of the control group and the groups treated with 100 mg/kg and 300 mg/kg body weight of *Zingiber officinale* extract over the 14-day experimental period. Body weight increased progressively in all groups throughout the study. However, the rate of weight gain was lower in the treatment groups than in the control group, with the greatest reduction observed in rats administered 300 mg/kg body weight of the extract.

The mean pituitary gland weight was 10.85 ± 0.04 mg in the control group, 7.30 ± 0.03 mg in the 300 mg/kg treatment group, and 5.19 ± 0.18 mg in the 100 mg/kg treatment group. Statistical analysis revealed significant differences ($P \leq 0.05$) between the control group and both treatment groups. The reduction in pituitary gland weight was more pronounced in rats administered 300 mg/kg body weight of *Zingiber officinale* extract.

3.2. Serum oxidative stress biomarkers

Table 2 presents the distribution and comparison of serum oxidative stress biomarkers, including malondialdehyde, CAT, GSH, and SOD, among the control group and the groups treated with 100 mg/kg and 300 mg/kg body weight of *Zingiber officinale* extract.

3.3. Pituitary gland histological changes

Figure 2 presents and compares the photomicrographs of the pituitary gland of adult female Wistar rats in the control and *Zingiber officinale* extract treated groups. The control showed normal pituitary histoarchitecture with well-preserved cellular morphology and normal tissue organisation. On the other hand, the rat treated with 100 mg/kg body weight of *Zingiber officinale* extract showed mild disruption of the normal cellular architecture with focal inflammatory cell infiltration. The rat treated with 300 mg/kg body weight of *Zingiber officinale* extract presented marked disruption of the normal cytoarchitecture, widespread cellular degeneration, and inflammatory cell infiltration.

Table 1: Mean body weight (grams) comparison across treatment groups

Days	Control	100mg/kg	300mg/kg	F-value	P-value
Day one	151.7 ± 1.7	175.7 ± 13.3	181.7 ± 13.6	2.983	0.116
Day three	155.7 ± 4.3	175.3 ± 15.1	181.0 ± 8.4	1.479	0.291
Day six	159.0 ± 8.2	177.7 ± 17.5	180.3 ± 8.3	3.132	0.107
Day nine	164.2 ± 5.1	176.3 ± 12.1	177.0 ± 15.0	26.950	0.001*
Day twelve	168.9 ± 7.5	175.2 ± 12.6	173.0 ± 10.1	17.664	0.061
Day fourteen	171.0 ± 6.4	174.9 ± 15.1	170.7 ± 11.5	28.361	0.000*

*Numbers represent Mean±S.E, *mean significant P≤0.05. Group A: Control. Group B: 100mg/kg of Zingiber officinale. Group C: 300mg/kg of Zingiber officinale.*

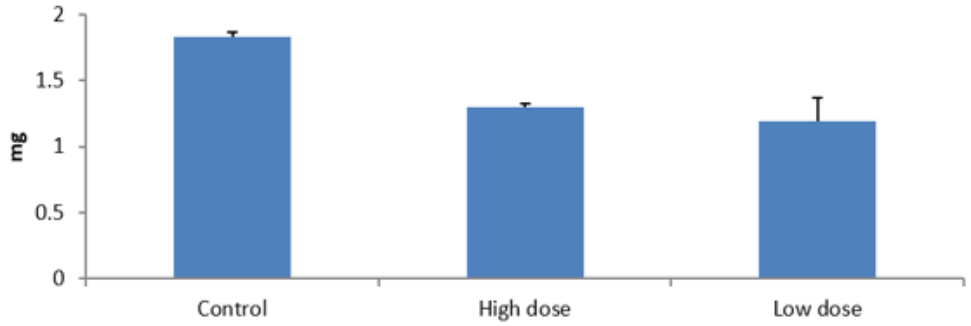


Figure 1: Comparison of pituitary gland weight in treatment groups
*Numbers represent Mean±S.E, *mean significant p≤0.05*.

Table 4.2: Antioxidant comparison across treatment groups

Antioxidants	Control	100mg/kg	300mg/kg	F-value	P-value
MDA (µmol/mg)	0.76 ± 0.35	1.19 ± 0.27	0.91 ± 0.30	1.390	0.331
Catalase (u/L	28.84 ± 3.04	75.15 ± 28.85	72.85 ± 50.22	1.141	0.391
GSH (u/L))	79.14 ± 102.05	27.10 ± 13.93	31.17 ± 6.98	0.866	0.475
SOD (u/mL)	143.89 ± 126.72	49.43 ± 11.29	59.11 ± 2.41	1.867	0.248

*Numbers represent Mean±S.E, *mean significant p≤0.05. MDA: Malondialdehyde. GSH: Reduces Glutathione. SOD: Superoxide Dismutase.*

4. DISCUSSION

The present study evaluated the dose-dependent effects of *Zingiber officinale* on body weight, pituitary gland weight, oxidative stress biomarkers, and pituitary histoarchitecture in adult female Wistar rats. The findings demonstrate that administration of *Zingiber officinale* influenced these parameters, with more pronounced effects observed at the higher dose of 300 mg/kg body weight.

Body weight increased progressively in all experimental groups during the 14-day study period. However, rats treated with *Zingiber officinale* exhibited a lower rate of weight gain than the control group, particularly those receiving the higher dose. These findings are consistent with those of Funmilola and Ayodeji [21], who reported reduced weight gain following chronic ginger administration, possibly through modulation of energy metabolism and appetite. In contrast, Li *et al*. [22] observed minimal changes in body weight following short-term administration, suggesting that the metabolic effects of ginger may depend on treatment dose, duration, and experimental conditions.

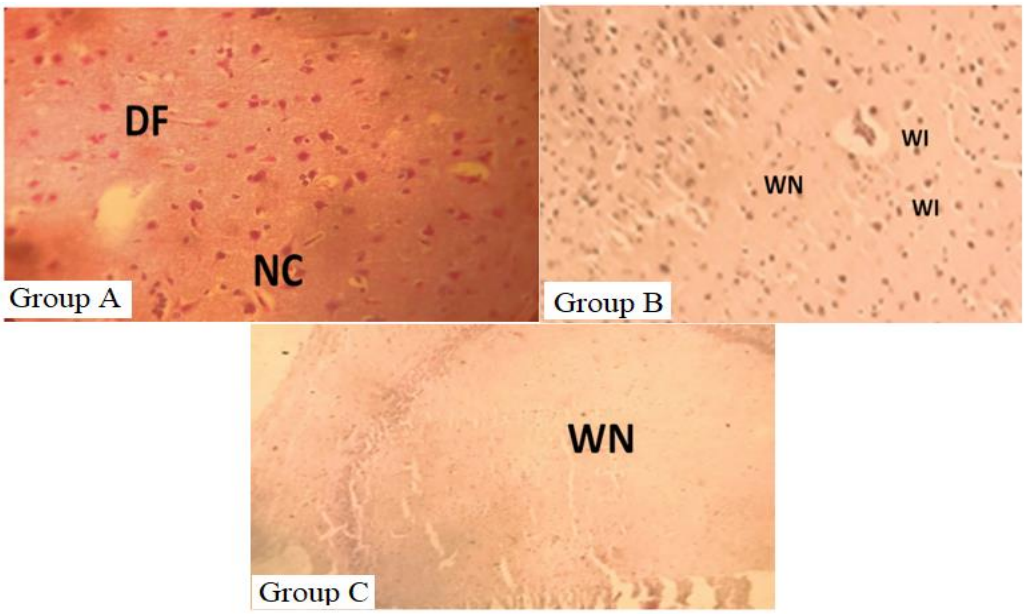


Figure 2. Histological features of the pituitary gland of Wistar rats following *Zingiber officinale* ingestion (H and E X400).

Pituitary gland weight was significantly reduced in both treatment groups compared with the control group, with the greatest reduction occurring in rats administered 300 mg/kg body weight of *Zingiber officinale*. This decrease may reflect structural alterations within the gland, as supported by the accompanying histological findings. Similar observations have been reported by Esegbue *et al*. [23], who demonstrated histological alterations in endocrine tissues following prolonged exposure to phytochemical-rich plant extracts. Conversely, Singh *et al*. [24] reported that moderate doses of ginger exerted either protective or negligible effects on endocrine tissues, suggesting that the biological response to ginger is influenced by dosage, duration of exposure, species, sex, and experimental design. Although pituitary weight may be associated with endocrine activity, hormonal assays were not performed in the present study; therefore, any effect on pituitary function should be interpreted with caution.

The effects of *Zingiber officinale* on oxidative stress biomarkers were variable. Catalase activity increased in both treatment groups, suggesting an enhancement of one aspect of the endogenous antioxidant defence system. This observation agrees with the findings of Ali *et al*. [25], who reported increased catalase activity following ginger supplementation. In contrast, GSH and SOD activities were lower in treated animals than in the control group. These findings may indicate increased utilisation of endogenous antioxidant reserves in response to phytochemical exposure or differential regulation of antioxidant pathways rather than a uniform enhancement of antioxidant status. Similar reductions in non-enzymatic antioxidants have been reported by Martins *et al*. [26] following exposure to certain phytochemical-rich plant extracts. Furthermore, although malondialdehyde concentrations were lower in the treated groups than in the control group, these differences were not statistically significant. Consequently, the antioxidant effects of *Zingiber officinale* observed in this study appear to be selective rather than universal.

Histological examination demonstrated normal pituitary architecture in the control group, whereas treatment with *Zingiber officinale* produced varying degrees of cytoarchitectural disruption and inflammatory cell infiltration. These alterations were more pronounced in

animals receiving 300 mg/kg body weight of the extract, suggesting a dose-dependent response. Comparable findings have been reported by Yang *et al*. [12], who observed cytotoxic alterations in endocrine tissues following exposure to high concentrations of plant-derived bioactive compounds. The histological changes observed in the present study are consistent with the reduction in pituitary gland weight and indicate that prolonged exposure to higher doses of *Zingiber officinale* may adversely affect pituitary tissue integrity.

Taken together, the present findings demonstrate a complex dose-dependent biological response to *Zingiber officinale*. Although its bioactive constituents, including gingerols and shogaols, are widely recognised for their antioxidant properties [27, 28], the present results indicate that increasing the dose does not necessarily confer greater biological benefit. The observed increase in catalase activity, together with reductions in GSH and SOD activities and the accompanying histological alterations, suggests that higher doses may simultaneously activate selected antioxidant mechanisms while inducing structural changes within the pituitary gland. These findings support the concept that the biological effects of phytochemicals are dose-dependent and may exhibit both protective and potentially adverse effects depending on the level and duration of exposure.

Overall, this study demonstrates that *Zingiber officinale* exerts dose-dependent effects on oxidative stress biomarkers and pituitary histoarchitecture in adult female Wistar rats. While the extract enhanced catalase activity, the reductions in GSH and SOD activities, together with the observed histological alterations, indicate that higher doses may compromise pituitary tissue integrity. Further studies incorporating hormonal assays, molecular biomarkers, and longer treatment durations are required to clarify the mechanisms underlying these effects and to establish the therapeutic safety profile of *Zingiber officinale*.

5. CONCLUSION

In this study, *Zingiber officinale* administration in adult female Wistar rats produced significant effects on body weight, pituitary gland weight, oxidative stress markers, and pituitary histology. Rats receiving ginger extracts, particularly at the higher dose of 300 mg/kg, showed slower weight gain compared to controls, suggesting potential metabolic modulation. Pituitary gland weight was significantly reduced in both treated groups, accompanied by histological changes, including cellular disruption and inflammation, indicating dose-dependent structural effects. Oxidative stress analysis revealed enhanced catalase activity alongside reduced GSH and SOD levels, highlighting selective modulation of antioxidant pathways.

These findings indicate that while *Zingiber officinale* possesses antioxidant properties, high doses may influence endocrine organ structure and function. The overall outcome underscores the importance of dose consideration in using ginger as a therapeutic agent. The study suggests that moderate, controlled administration of *Zingiber officinale* may offer antioxidant benefits without adversely affecting pituitary structure, providing a foundation for future research on safe and effective dosage in managing oxidative stress-related conditions.

Data availability statement: The data supporting the findings of this study are available from the corresponding author upon reasonable request. All datasets generated and analyzed during this research are stored in the Department of Anatomy, Ambrose Alli University, Ekpoma, Edo State, Nigeria.

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Conflict of interest: The authors declare that there are no conflicts of interest regarding the publication of this manuscript.

Ethics approval: Ethical approval for this study was obtained from the Institutional Animal Ethics Committee of Ambrose Alli University, Ekpoma, Edo State, Nigeria. All animal handling and experimental procedures were conducted in accordance with international guidelines for the care and use of laboratory animals.

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